

Longitudinal forecasting from the current clinical state

Historical Qwen3.5-0.8B pilot | matched full-token and eight learned-query-slot forecasters

Current CXR
available at prediction

Copy Current
copies current report

Full-token baseline
0.8B matched forecast model

Eight-slot pilot
0.8B; final-language-layer slots

Observed follow-up
REFERENCE ONLY

a) Label onset 0 → 1 | Pleural effusion

Input horizon: >24-72 h · Observed gap: 41.2 h



Current label: absent
PA view · current image + report + pre-image clinical history

Future effusion label

Absent

No continuous probability

Current report excerpt

There is no evidence of complicating pleural effusions.

Future effusion probability

56.6%

Generated report excerpt

[No effusion-specific sentence in generated report.]

Report label: not mentioned

Future effusion probability

54.3%

Generated report excerpt

[No effusion-specific sentence in generated report.]

Report label: not mentioned



Reference label: present

b) Label resolution 1 → 0 | Pleural effusion

Input horizon: 6-24 h · Observed gap: 16.9 h



Current label: present
AP view · current image + report + pre-image clinical history

Future effusion label

Present

No continuous probability

Current report excerpt

[...] and worsening congestive heart failure with increasing perihilar edema and persistent small right pleural effusion.

Future effusion probability

81.6%

Generated report excerpt

No larger pleural effusions.

Report label: absent

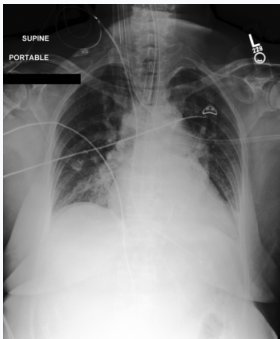
Future effusion probability

79.3%

Generated report excerpt

Small bilateral pleural effusions are presumed.

Report label: present



Reference label: absent

c) Label persistence 1 → 1 | Pleural effusion

Input horizon: 6-24 h · Observed gap: 22.8 h



Current label: present
AP view · current image + report + pre-image clinical history

Future effusion label

Present

No continuous probability

Current report excerpt

increase in moderate right pleural effusion.

Future effusion probability

97.3%

Generated report excerpt

No larger pleural effusions.

Report label: absent

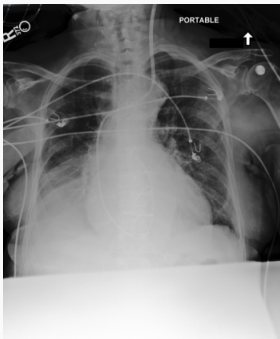
Future effusion probability

96.9%

Generated report excerpt

The bilateral pleural effusions are constant in extent and severity.

Report label: present



Reference label: present

Selection: fixed finding; first eligible pair by sorted ID for each reference-label transition; distinct patients; explicit report-text consistency screen.
CheXbert report labels are not clinician-adjudicated changes. Probabilities come from finding heads; report labels can disagree. Follow-up images are never model inputs.
Pilot review artifact: these outputs do not evaluate the pending 9B run or the proposed multi-depth fusion/vision 4+4 architecture.